Supplementary Material

S1. Cohort Flow and Missingness

Of 267 subjects in the spine-generic dataset (release r20231212), 203 were healthy controls (HC) and 64 had pathology (62 mild cervical compression; 2 degenerative cervical myelopathy). Both cohorts were pre-specified prior to analysis.

Missingness patterns. Among HC subjects, 15 (7.4%) were missing one or more biomarkers: MTR ($n = 9$), MTsat ($n = 13$), and each of FA, MD, AD and RD ($n = 5$). Among ALL subjects, 21 (7.9%) were incomplete: MTR ($n = 13$), MTsat ($n = 18$), and each of FA, MD, AD and RD ($n = 7$). The complete-case HC cohort comprised $N = 188$ and the complete-case ALL cohort $N = 246$.

Missingness mechanism. No subject was excluded because of a quality-control score. Every missing value was traced to the acquisition or aggregation stage, and all 21 incomplete ALL-cohort subjects were attributed as follows: MT series not acquired at the site (8 subjects, comprising every subject at two sites); MT-off series with the second flip angle absent, affecting MTsat only (5 subjects); diffusion series stored as multiple runs under a file-naming convention the pipeline did not recognise, so the input was skipped (3 subjects); vertebral labelling placed the C2–C3 level outside the field of view of the MT acquisition, so extraction returned no voxels; diffusion metrics were lost for the same reason in two of these subjects and through the identically-zero rule in the third (3 subjects; see S2); a metric map identically zero within a non-empty cord mask and nulled by the aggregation scripts (1 subject, MTR); and site-level processing not executed for that subject (1 subject, MT and diffusion). Subjects are assigned to a single category except for one subject with two contributing mechanisms, counted under the first.

Because several mechanisms operate through the site acquisition protocol, incompleteness clustered by site—three sites were affected in every one of their subjects—and varied across vendors (GE 10.8%, 4/37; Philips 6.0%, 3/50; Siemens 7.8%, 14/180). Incomplete subjects were older than complete subjects (median 34.0 vs. 28.5 years, Mann–Whitney $p = 0.015$) but did not differ in sex (χ^2 $p = 0.880$). Missingness was therefore not completely at random with respect to age, site or vendor. All analyses are complete-case, the missingness mechanism was not modelled, and this is stated as a limitation in the main text.

Table S1. Demographic characteristics by vendor (ALL cohort, $N = 246$).

Characteristic	**GE** ($n = 33$)	**Philips** ($n = 47$)	**Siemens** ($n = 166$)	**Total** ($n = 246$)	p
Sites (n)	5	8	27	40	—
Age (years)	28.0 ± 5.2	29.4 ± 4.7	30.0 ± 6.3	29.6 ± 5.9	0.193
Sex (% female)	48.5	46.8	47.0	47.2	0.986
Pathology (HC/MC/DCM)	25/8/0	33/13/1	130/35/1	188/56/2	0.661

S2. Biomarker Extraction and Quality Control

All biomarkers were extracted using the Spinal Cord Toolbox (`https://spinalcordtoolbox.com`) following the spine-generic processing pipeline documentation (`https://spine-generic.readthedocs.io`), with the differences listed in Table S2a below. The installed build reports version `7.4.dev0`, which is not a numbered release; the exact source snapshot was the master branch at git commit `93923f5e363d9400fbadab07459f125a08b21ad8` (2 June 2026, sixteen commits after the 7.3 release of 8 May 2026). All segmentations, vertebral labels and metric maps were generated de novo for this study; the manual disc labels and segmentations shipped in the dataset's `derivatives/labels` directory were not used, so that every subject was processed by one uniform, fully automatic pipeline. Per-subject QC reports (segmentation and labelling overlays) were generated with `sct_qc` for every subject and contrast and are available on request.

Table S2a. Step-by-step comparison of the present pipeline with the spine-generic reference pipeline (`process_data.sh`, `https://github.com/spine-generic/spine-generic`).

Step	Reference pipeline (process_data.sh)	Present pipeline
Software	SCT available at data release (2021–2023 era)	SCT `7.4.dev0`, master commit `93923f5e` (2026-06-02)
Cord segmentation	`sct_deepseg_sc` (legacy per-contrast CNN); manual segmentations used where shipped	`sct_deepseg spinalcord` (contrast-agnostic nnUNet), fully automatic for all subjects
GM segmentation	`sct_deepseg_gm` (legacy)	`sct_deepseg graymatter` (nnUNet, model release r20250420)
Vertebral labelling	`sct_label_vertebrae` on T1w, using the shipped manual disc labels (`-discfile`) where available; PAM50 levels propagated via template registration	`sct_label_vertebrae` on T2w, fully automatic (no manual disc labels); T2w labels registered into the MT and DWI spaces (audit in S2.2)
T2w cord CSA	C2–C3 (`-vert 2:3`; a C3–C4 variant also computed)	C2–C3 (`-vert 2:3`) — unchanged
GM-CSA ($T2^*$)	C3–C4 (`-vert 3:4`, angle correction off)	Mean over every segmented cervical slice (no level restriction; $T2^*$ coverage shorter than the levels resolved on T2w)
MTR / MTsat	White-matter PAM50 atlas, C2–C5 (`-vert 2:5`)	Whole-cord mask, C2–C3 (`-vert 2:3 -method wa`)
DTI metrics	FA, MD, RD in the white-matter atlas, C2–C5; AD not extracted	FA, MD, AD, RD in the whole-cord mask, C2–C3

The common C2–C3 window was chosen so that the seven level-restricted biomarkers sample identical anatomy; the benchmark conclusions concern method rankings on the same data rather than absolute biomarker values, and the window applies identically to all methods and both cohorts.

Processing steps:

- **Vertebral labelling:** `sct_label_vertebrae` on the T2w image. This command registers the PAM50 template to the native space and propagates disc labels; each axial slice was then assigned to the vertebral level spanning it, and slices at levels 2 and 3 formed the C2–C3 window.
- **T2w-CSA:** `sct_deepseg spinalcord` on the T2w image, followed by `sct_process_segmentatic` with `-vert 2:3`.
- **MTR/MTsat:** Computed from MT-on and MT-off acquisitions (`sct_compute_mtr`, `sct_compute_mtsat`) and averaged over the C2–C3 window with `sct_extract_metric -vert 2:3 -method wa`, after registration of the T2w vertebral labels into the MT space.
- **DTI metrics:** `sct_dmri_moco` and `sct_dmri_compute_dti`, then averaged over the C2–C3 window with `sct_extract_metric -vert 2:3 -method wa` after the

same registration into the diffusion space.

- **GM-CSA:** `sct_deepseg graymatter` (nnUNet model, release r20250420) on the T2*-weighted image. T2*-derived cord, grey-matter and white-matter CSA were averaged over every segmented slice and were not restricted to C2–C3, because the T2* acquisition covers a shorter craniocaudal range than the vertebral levels resolved on T2w.

Quality control. No subject was excluded because of a quality-control score. The only automated exclusions were two zero-value rules applied during aggregation rather than at the segmentation or model-fitting stage: MTR (and hence MTsat, plus their standard deviations) set to missing when MTR was identically zero within a non-empty cord mask, and all four diffusion metrics set to missing when FA was identically zero within a non-empty cord mask. These two rules account for one subject each. All other missing values originated in the acquisition or aggregation stages listed in S1.

S2.2 Audit of vertebral labelling. Because every C2–C3 metric depends on the output of `sct_label_vertebrae`, we screened all subjects for displacement of the labelling chain. The cord segmentation extends cranially to the medulla, whereas the labelling chain terminates at its most cranial disc; the distance between the two therefore measures the unlabelled medullary segment, and was 9.6 mm (median) in the cohort. Fifteen subjects–all in the HC cohort, 7.4% of the 203 HC subjects–exceeded 18 mm and, on visual review, showed the labelling chain placed at least one vertebral level too caudal. The affected subjects clustered within sites—two sites contributed two subjects each and eleven sites one each—which is consistent with a site-level rather than random origin.

For these 15 subjects the C2–C3 window was relocated interactively: the most cranial tip of the imaged cord was placed by hand, and the whole C2–C3 window was then translated rigidly, with its width fixed at that subject's own C2 and C3 height (28–34 mm). All seven level-restricted biomarkers were re-extracted (GM-CSA is unaffected, being averaged over the whole segmented cord). Of the 105 resulting values, 81 could be re-extracted. T2w-CSA was recovered for all 15 subjects. For three subjects (fslAchieva01, pavia05, tehranS06) the relocated C2–C3 level lay outside the field of view of both the MT and the diffusion acquisitions, so no voxels could be extracted at all; these three were already missing six biomarkers each in the primary analysis, so the complete-case sample sizes are unaffected. For one further subject (oxfordFmrib03) only two diffusion slices (10 mm) fell inside the field of view, below a three-slice minimum, so its six MT and diffusion metrics were not replaced. Changes in T2w-CSA ranged from -13.1 to $+26.2$ mm^2, and translating the window by ± 8 slices moved T2w-CSA by 1.1–6.8 mm^2, so the relocated values are not sensitive to the precision of the manual placement.

Sensitivity of the reported findings. Replacing the 81 re-extracted values in the complete-case cohorts changed the baseline vendor η^2 (mean over eight biomarkers) from 0.3147 to 0.3090 in HC ($N = 188$) and from 0.3360 to 0.3309 in ALL ($N = 246$);

PERMANOVA remained $p = 0.001$ in both cohorts. An independent check supports the relocation: the ratio of T2w-CSA to the $T2^*$-derived cord CSA, which is unaffected by the mislabelling because it is not level-restricted, lay outside the cohort 1st–99th percentile range (0.841–1.031) for 5 of the 15 subjects in the primary analysis and for none of them after relocation.

The primary analysis reported in the main text retains the values produced by the automated pipeline. The relocated values are reported here as a sensitivity analysis rather than adopted as primary, for three reasons: the interactive placement is not an anatomical gold standard; the defect is irrecoverable in the three subjects whose relocated C2–C3 level falls outside the MT and diffusion field of view; and the effect on the reported statistics is small.

S3. Harmonization Methods: Mathematical Details

S3.1 ComBat (per-metric). For each biomarker y_{ijv} for subject i at site v:

$$y_{ijv}^{\text{harm}} = \frac{y_{ijv} - \hat{\alpha}_v - X_{ij}\hat{\beta}}{\hat{\gamma}_v} + X_{ij}\hat{\beta}$$

where $\hat{\alpha}_v$ and $\hat{\gamma}_v$ are vendor-specific location and scale parameters estimated with empirical Bayes shrinkage, and X_{ij} is the design matrix of biological covariates (age, sex). This is the standard ComBat formulation [8,9] applied independently to each biomarker.

S3.2 ComBat-joint. All eight biomarkers are modeled jointly with a shared vendor parameter estimated via joint empirical Bayes (`eb=True` in neuroCombat). The mathematical form is identical to S3.1 but $\hat{\alpha}_v$ and $\hat{\gamma}_v$ are estimated simultaneously across biomarkers, preserving cross-metric covariance structure.

S3.3 CovBat. After ComBat-joint harmonization, the residual matrix E (subjects $\times$ biomarkers) is decomposed via PCA: $E = ZW^T$, where Z is the PC score matrix and W is the loading matrix. PCs explaining at least 95% of the residual variance were retained ($K = 5$ of 8 in both cohorts; see Supplementary Fig. S4 for the resulting covariance structure). Each PC k is then harmonized independently: $Z_k^{\text{harm}} = (Z_k - \hat{\alpha}_{vk})/\hat{\gamma}_{vk}$, where $\hat{\alpha}_{vk}$ and $\hat{\gamma}_{vk}$ are vendor-specific parameters for PC k. The harmonized data are reconstructed as $Y^{\text{harm}} = Y^{\text{ComBat-joint}} + Z^{\text{harm}}W^T$.

S3.4 RELIEF. After ComBat-joint harmonization, residuals undergo ICA decomposition: $E = SA^T$, where S is the independent component score matrix and A is the mixing matrix. For each component c, a one-way ANOVA tests vendor association. Components with Benjamini-Hochberg FDR-controlled $q < 0.05$ are zeroed ($S_c \leftarrow 0$). The remaining components are recombined: $E^{\text{RELIEF}} = S^{\text{cleaned}}A^T$.

S3.5 LME/FE. For each biomarker:

$$y_{ijv} = \beta_0 + \beta_1 \mathrm{Age}_i + \beta_2 \mathrm{Sex}_i + u_v + \varepsilon_{ijv}$$

where $u_v \sim \mathcal{N}(0, \sigma_u^2)$ is the vendor-specific random intercept. Harmonized values: $y_{ijv}^{\mathrm{harm}} = y_{ijv} - \hat{u}_v$ when LME converges; $y_{ijv}^{\mathrm{harm}} = y_{ijv} - \bar{y}_v + \bar{y}$ (OLS centering) when the random-effects covariance matrix is singular (FE fallback).

S4. Evaluation Metrics: Computational Details

S4.1 Vendor η^2. One-way ANOVA: $\eta^2 = \mathrm{SS}_{\mathrm{vendor}}/\mathrm{SS}_{\mathrm{total}}$. Reported as mean across 8 biomarkers.

S4.2 PERMANOVA. Euclidean distance matrix on the 8-dimensional biomarker space after per-dimension z-scoring. 9,999 permutations of vendor labels. $R^2 = \mathrm{SS}_{\mathrm{vendor}}/\mathrm{SS}_{\mathrm{total}}$ from the distance-based pseudo-F statistic [20]. Implementation: custom NumPy code (pseudo-F and permutation loop written in-house; no third-party PERMANOVA package). On z-scored features with Euclidean distance, the multivariate R^2 equals the mean of the per-dimension η^2 computed on the same subjects. Because all five methods and the unharmonised baseline are fitted and evaluated on the same unified complete-case sample (HC $N = 188$, ALL $N = 246$), this identity holds exactly for every method, so univariate and multivariate summaries are directly comparable.

S4.3 Biological R^2. Type-II semi-partial R^2 from linear models: biomarker $\sim$ Age + Sex. Reported as mean across biomarkers.

S4.4 Subject-level preservation. For each biomarker, the Pearson correlation across subjects between the pre- and post-harmonization values—i.e., preservation of the subject ranking on that biomarker—averaged over the eight biomarkers:

$$r = \mathrm{mean}_b\big[\mathrm{corr}_i\big(y_{ib}^{\mathrm{pre}}, y_{ib}^{\mathrm{post}}\big)\big],$$

where b indexes biomarkers and i subjects. This definition is consistent with the $\sqrt{1-\eta^2}$ ceiling argument in S6.

S4.5 Unified complete-case analysis sample. All results in the main text and this supplement are computed on a single pre-specified complete-case sample per cohort: HC $N = 188$ and ALL $N = 246$ (all eight biomarkers plus age, sex, and vendor observed). ComBat and LME/FE were refitted on this unified sample (the ComBat and LME/FE columns of every table therefore derive from these refits); ComBat-joint, CovBat, and RELIEF are complete-case by construction because they require the full joint biomarker panel. This unified definition replaces the earlier per-metric maximum-available-sample evaluation and removes any sample mismatch between univariate and multivariate endpoints.

S4.6 Within-resample paired bootstrap. To compare $r_{\text{pre,post}}$ between method pairs, each stratified bootstrap resample ($B = 2{,}000$, resampling subjects within vendor strata) was passed to *all* methods simultaneously; the per-resample differences $r_A - r_B$ form the null distribution for the pairwise test. This pairing removes between-resample variance (the source of the overlapping marginal CIs) and directly targets the pairwise contrast. Results: in HC, 8 of 10 method pairs were significantly separated (all $p \leq 0.003$; ComBat vs. CovBat $p = 0.070$ and ComBat vs. RELIEF $p = 0.694$ were not); in ALL, 9 of 10 pairs were separated (ComBat vs. RELIEF $p = 0.403$ was not). LME/FE was significantly highest and CovBat significantly lowest in both cohorts. Full results in Supplementary Table S7.

Table S7. Within-resample paired bootstrap comparison of subject-level preservation ($r_{\text{pre,post}}$) between all method pairs ($B = 2{,}000$, stratified by vendor; complete-case sample). $\Delta r = r_A - r_B$; CI = 95% percentile interval of the per-resample differences; p = two-sided bootstrap p-value. HC cohort above, ALL cohort below.

Cohort	Comparison	Δr	95% CI	p	Significant
HC	ComBat vs. ComBat-joint	−0.017	[−0.020, −0.014]	0.0005	Yes
	ComBat vs. CovBat	0.007	[−0.000, 0.015]	0.070	No
	ComBat vs. RELIEF	−0.001	[−0.007, 0.005]	0.694	No
	ComBat vs. LME/FE	−0.027	[−0.033, −0.020]	0.0005	Yes
	ComBat-joint vs. CovBat	0.024	[0.017, 0.032]	0.0005	Yes
	ComBat-joint vs. RELIEF	0.016	[0.010, 0.022]	0.0005	Yes
	ComBat-joint vs. LME/FE	−0.010	[−0.015, −0.004]	0.003	Yes
	CovBat vs. RELIEF	−0.009	[−0.015, −0.003]	0.002	Yes
	CovBat vs. LME/FE	−0.034	[−0.043, −0.025]	0.0005	Yes
	RELIEF vs. LME/FE	−0.026	[−0.033, −0.018]	0.0005	Yes
ALL	ComBat vs. ComBat-joint	−0.013	[−0.015, −0.011]	0.0005	Yes
	ComBat vs. CovBat	0.011	[0.005, 0.018]	0.001	Yes
	ComBat vs. RELIEF	0.002	[−0.003, 0.008]	0.403	No
	ComBat vs. LME/FE	−0.027	[−0.033, −0.020]	0.0005	Yes
	ComBat-joint vs. CovBat	0.024	[0.018, 0.031]	0.0005	Yes
	ComBat-joint vs. RELIEF	0.015	[0.010, 0.020]	0.0005	Yes
	ComBat-joint vs. LME/FE	−0.014	[−0.019, −0.008]	0.0005	Yes
	CovBat vs. RELIEF	−0.009	[−0.015, −0.004]	0.001	Yes
	CovBat vs. LME/FE	−0.038	[−0.046, −0.029]	0.0005	Yes
	RELIEF vs. LME/FE	−0.029	[−0.035, −0.022]	0.0005	Yes

S5. LOVO External Validation: Protocol

Training. For each method, harmonization models were fitted on subjects from two vendors (training set). The third vendor was entirely held out as test set.

Transformation of unseen vendors:

- **ComBat/ComBat-joint:** Test-vendor biomarker values were z-scored to the training-vendor harmonized distribution: $y_{\text{test}}^{\text{harm}} = (y_{\text{test}} - \mu_{\text{test}}) \times (\sigma_{\text{train}}^{\text{harm}}/\sigma_{\text{test}}) + \mu_{\text{train}}^{\text{harm}}$.
- **CovBat:** Test-vendor residuals were projected onto training PCA space; PC scores were z-score matched to training PC-score distribution.
- **RELIEF:** Test-vendor residuals were projected onto training ICA space via $\text{inv}(A^T)$; batch-flagged components were zeroed using the training-vendor flag set.
- **LME/FE:** Test-vendor offset set to 0; test-vendor values were carried through essentially untransformed (raw values, no vendor or covariate adjustment).

Evaluation. The same three endpoints were computed after pooling the harmonised training-vendor and held-out test-vendor data within each fold, then averaged across the 3 rotation folds. Because the transformation applied to the test vendor is per-biomarker affine (location–scale) for ComBat, ComBat-joint, and LME/FE, the test-vendor pre/post correlation equals 1.0 by construction for these methods; their preservation statistics therefore reflect the pooled sample rather than the unseen vendor in isolation. Similarly, the LOVO/internal η^2 ratio below 1 for ComBat-joint reflects the pooled evaluation sample rather than genuine improvement on unseen vendors. For CovBat and RELIEF, projection onto the training PCA/ICA basis mixes biomarker dimensions, and the test-vendor pre/post correlation fell below 1 ($r = 0.85$–0.97 across folds), providing a genuine held-out-vendor distortion measure (Table S6).

Table S6. Test-vendor-only evaluation metrics by held-out vendor and method (mean across 8 biomarkers; complete-case training sample $N = 188$). |shift|: mean absolute shift of the held-out-vendor mean relative to the pooled training distribution, in training-SD units; var ratio: held-out-vendor to training variance ratio; centroid shift: multivariate centroid shift in pooled SD units; $r_{\text{pre,post}}$: test-vendor pre/post correlation. ComBat and ComBat-joint force the test-vendor distribution onto the training template (shift exactly 0 by construction); CovBat leaves small residual shifts; RELIEF and LME/FE leave the held-out vendor shifted relative to training.

Held-out	Method	|shift|	var ratio	centroid shift	$r_{\text{pre,post}}$ (test)
GE	Original	1.46	0.98	5.68	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.03	0.94	0.12	0.851
	RELIEF	1.64	1.12	5.97	0.925
	LME/FE	1.62	1.08	5.92	1.000
Philips	Original	0.80	0.83	2.58	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.03	1.00	0.10	0.949
	RELIEF	0.85	0.97	2.69	0.973
	LME/FE	0.92	0.99	3.14	1.000
Siemens	Original	1.05	0.93	3.37	1.000
	ComBat	0.00	1.00	0.00	1.000
	ComBat-joint	0.00	1.00	0.00	1.000
	CovBat	0.05	0.96	0.16	0.911
	RELIEF	1.42	1.24	4.54	0.960
	LME/FE	1.39	1.13	4.51	1.000

S6. Design C Subsample Analyses (Reduced Imbalance and Equal-n)

To isolate the effect of vendor imbalance, two subsample designs were analysed in the HC cohort (20 random seeds each, all five methods refitted per subset using the same canonical implementations as the main analysis; subject-level preservation is reported as the pooled $r_{\text{pre,post}}$, the same metric as Table 3 of the main text). The reduced-imbalance design restricted Siemens to 40 randomly drawn subjects per seed while retaining all GE and Philips subjects (GE 25, Philips 33, Siemens 40; $n \approx 98$ per subset; residual imbalance max / min = 1.60 vs. 5.2 in the full analysis). The true-equal-n sensitivity design drew 25 subjects per vendor ($n = 75$; imbalance ratio = 1.00). Vendor η^2, PERMANOVA R^2, pooled and within-vendor subject-level preservation, and age-signal recovery were recomputed per seed.

In the reduced-imbalance design the pooled-r preservation hierarchy was retained (LME/FE $r = 0.779 \pm 0.011$, ComBat-joint 0.773 ± 0.014, RELIEF 0.760 ± 0.015, CovBat 0.749 ± 0.016, ComBat 0.745 ± 0.015; the only rank change vs. the full analysis was the adjacent exchange of ComBat and CovBat, a pair not significantly separated

in the main analysis), and the true-equal-n design gave concordant results (LME/FE 0.772 ± 0.015, ComBat-joint 0.768 ± 0.016, RELIEF 0.754 ± 0.014, CovBat 0.743 ± 0.017, ComBat 0.732 ± 0.017). Absolute values were lower than in the full cohort (0.782–0.816) because the subsamples carried a higher baseline vendor-variance fraction: the measured unharmonised η^2 was 0.374 (range 0.343–0.401 across seeds) in the reduced-imbalance design and 0.389 (range 0.337–0.431) in the true-equal-n design, versus 0.315 in the full analysis. Because pooled r after a location–scale correction cannot exceed $\sqrt{1-\eta^2}$, these baselines imply ceilings of approximately 0.79 and 0.78; every per-method mean fell below its design's ceiling, and in a per-seed check the LME/FE pooled r never exceeded the per-seed ceiling $\sqrt{1-\eta^2}$ in any of the 20 seeds of either design (smallest margins 0.007 and 0.004). Within-vendor subject rankings were nonetheless preserved near-perfectly by every method in both designs (mean within-vendor $r \geq 0.97$), indicating that the pooled-r gradient reflects differential removal of between-vendor variance rather than degradation of within-vendor subject rankings. Results are shown in Supplementary Fig. S2.

S7. Supplementary Figures

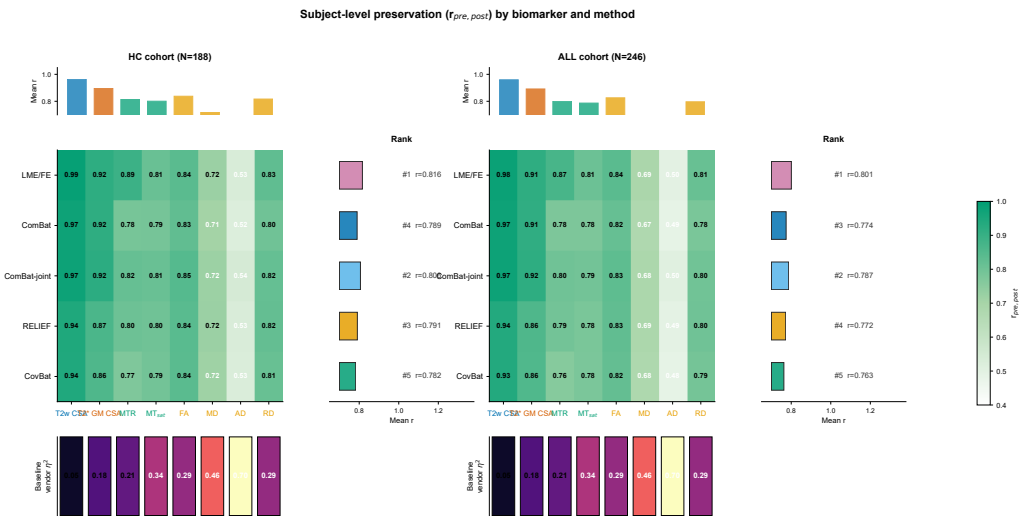


Figure 1: Subject-level preservation ($r_{\text{pre,post}}$) by method. Marginal distributions of per-subject Pearson correlations between pre- and post-harmonization 8-dimensional biomarker vectors. Higher values indicate better preservation of subject ranking. LME/FE shows highest preservation ($r = 0.82$); CovBat lowest ($r = 0.78$).

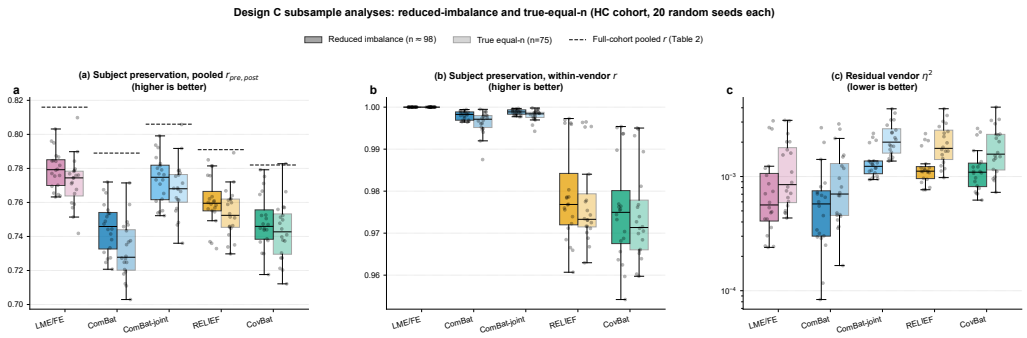


Figure 2: Design C subsample analyses: reduced-imbalance and true-equal-n designs (HC cohort, 20 random seeds each; all methods refitted per subset). **(a)** Pooled subject-level preservation ($r_{\text{pre,post}}$, the main-text Table 3 metric): reduced-imbalance subsample (Siemens restricted to 40 subjects per seed, all GE and Philips subjects retained; $n \approx 98$ per subset; imbalance ratio 1.60) and true-equal-n subsample (25 subjects per vendor, $n = 75$, ratio 1.00); dashed lines mark the full-cohort reference values. **(b)** Within-vendor subject-ranking preservation (Pearson r computed within each vendor group; equal to 1 for pure location-shift methods by construction). **(c)** Vendor effect size (η^2) on logarithmic scale; the dashed line marks the unharmonised subsample baseline (mean $\eta^2 \approx$ 0.37–0.39). The pooled-r preservation hierarchy was retained in both designs (reduced imbalance: LME/FE 0.779 ± 0.011, ComBat-joint 0.773 ± 0.014, RELIEF 0.760 ± 0.015, CovBat 0.749 ± 0.016, ComBat 0.745 ± 0.015; equal-n: LME/FE 0.772 ± 0.015, ComBat-family 0.732–0.768), with absolute values below the full cohort (0.782–0.816) because the subsamples carry a higher baseline vendor-variance fraction; within-vendor rankings were preserved near-perfectly by every method (mean $r \geq 0.97$).

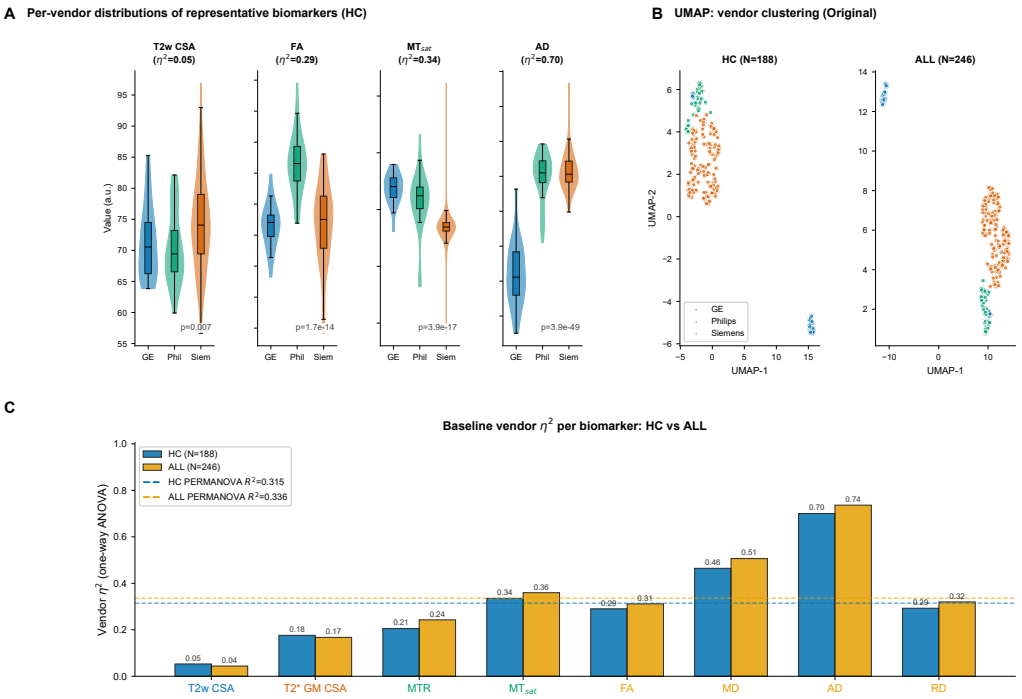


Figure 3: Baseline vendor effects in spinal cord qMRI biomarkers. **(a)** Per-vendor distributions of four representative biomarkers in the HC cohort (T2w-CSA $\eta^2 = 0.05$; FA 0.29; MTsat 0.34; AD 0.70; one-way ANOVA p values shown). **(b)** UMAP embedding of the 8-dimensional biomarker space (Original, unharmonised data) coloured by vendor, for the HC ($N = 188$) and ALL ($N = 246$) cohorts. **(c)** Per-biomarker baseline vendor η^2 in the HC and ALL cohorts; mean $\eta^2 = 0.31$ (HC) and 0.34 (ALL); PERMANOVA $R^2 = 0.315$ (HC) and 0.336 (ALL), both $p = 1 \times 10^{-4}$ (permutation lower bound with 9,999 permutations).

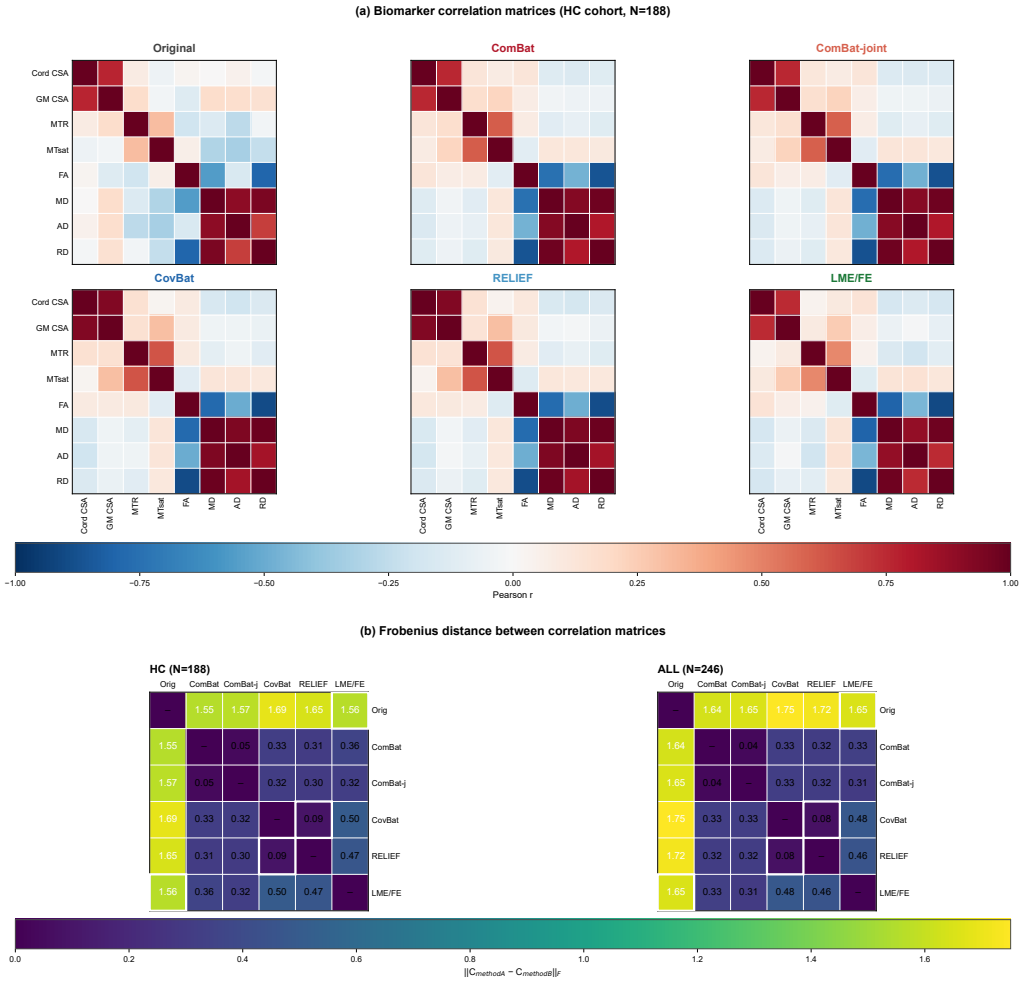


Figure 4: Cross-metric covariance structure preservation. **(a)** Pearson correlation matrices of the eight biomarkers (HC cohort, $N = 188$) before and after each method. **(b)** Frobenius distance between the correlation matrices of each pair of methods (lower = more similar covariance structure). Expressed relative to the total Frobenius norm of the original correlation matrix ($\|R_{\text{orig}}\|_F = 4.08$ in HC, 4.07 in ALL), all methods displaced the covariance structure by 38.1–41.3% in HC (ComBat 38.1%, LME/FE 38.3%, ComBat-joint 38.4%, RELIEF 40.3%, CovBat 41.3%; 40.2–43.0% in ALL: ComBat 40.2%, ComBat-joint 40.5%, LME/FE 40.6%, RELIEF 42.3%, CovBat 43.0%), so covariance preservation holds only in relative terms. CovBat and RELIEF are the closest to each other (HC: 0.094, ALL: 0.078), indicating near-identical covariance structure despite the additional ICA round-trip in RELIEF. ComBat and ComBat-joint are likewise nearly identical to each other (HC: 0.048, ALL: 0.036), showing that joint empirical Bayes has limited impact on the covariance structure beyond the per-metric ComBat baseline. Across all ten method pairs the remaining eight values lay between 0.30 and 0.50 in both cohorts. The location–scale methods (ComBat HC: 1.55, ALL: 1.64; LME/FE HC: 1.56, ALL: 1.65) remain marginally closer to the original than the covariance-projection methods (CovBat HC: 1.69, ALL: 1.75; RELIEF HC: 1.65, ALL: 1.72).

Table S8. Per-vendor one-vs-rest AUC (ALL cohort, $N = 246$) after fold-internal harmonisation. Hanley–McNeil 95% CI based on the point estimate; vendor labels are known to the harmonisation step (training and test) but the random forest is blind to them.

Method	GE AUC	95% CI	Philips AUC	95% CI	Siemens AUC	95% CI
Original	0.980	[0.947, 1.014]	0.987	[0.964, 1.010]	0.993	[0.984, 1.002]
ComBat	0.803	[0.709, 0.896]	0.666	[0.575, 0.758]	0.722	[0.658, 0.786]
ComBat-joint	0.766	[0.668, 0.865]	0.669	[0.578, 0.761]	0.729	[0.666, 0.792]
CovBat	0.585	[0.476, 0.693]	0.698	[0.609, 0.788]	0.669	[0.600, 0.738]
RELIEF	0.725	[0.622, 0.828]	0.691	[0.601, 0.781]	0.738	[0.676, 0.800]
LME/FE	0.907	[0.838, 0.977]	0.722	[0.634, 0.810]	0.849	[0.802, 0.895]

S8. Supplementary Data

The following data files are provided:

- `demographics_completecase.csv`: Complete-case analysis sample demographics by vendor and cohort (Supplementary Table S1)
- `lovo_summary.csv`: Leave-one-vendor-out validation results (18 rows)
- `lovo_testvendor.csv`: Leave-one-vendor-out held-out (test) vendor shifts and correlations per fold, method, and biomarker (162 rows; Supplementary Table S6)
- `corr_matrices_HC.csv` / `corr_matrices_ALL.csv`: Full 8×8 Pearson correlation matrices for all six methods (Supplementary Fig. S4)
- `frobenius_distance_HC.csv` / `frobenius_distance_ALL.csv`: Pairwise Frobenius distances between method correlation matrices (Supplementary Fig. S4)
- `comparison_long.csv`: Per-biomarker evaluation metrics (vendor η^2, age R^2, sex R^2, $r_{\text{pre,post}}$) for all methods and both cohorts on the unified complete-case sample (96 rows)
- `cc_summary.csv`: Method-level summary (mean/median/min/max) of the per-biomarker metrics in `comparison_long.csv`
- `permanova_results.csv`: PERMANOVA results (HC and ALL cohorts)
- `bootstrap_r_ci_cc.csv` / `bootstrap_eta2_baseline_cc.csv`: Bootstrap 95% confidence intervals for $r_{\text{pre,post}}$ and baseline η^2 ($B = 2{,}000$, stratified by vendor)
- `bootstrap_paired_r.csv`: Paired bootstrap tests of $r_{\text{pre,post}}$ differences between method pairs (20 rows; Supplementary Table S7)
- `ml_leakfree_summary.csv`: Fold-internal machine-learnability summary (vendor, pathology, and sex AUC by method and cohort)
- `ml_leakfree_vendorclass.csv`: Per-vendor one-vs-rest AUC after fold-internal harmonisation (Supplementary Table S8)
- `ml_leakfree_paired.csv`: Paired comparison of pathology AUC (Original vs each

method, within-resample)

- `lovo_permanova_z.csv`: Z-scored PERMANOVA results for the LOVO pooled evaluation sample
- `r1_age_confound.csv`: Age-confounding sensitivity analysis of the pathology-label classification (Supplementary Material S9)
- `r1_site_harmonization.csv`: ComBat with site and scanner-model batch variables (Supplementary Material S10)

The analysis code is available at `https://github.com/wudengke2010/spine-generic-harmonization`.

S9. Age Confounding of the Radiological Pathology Labels

The pathology labels are radiological assessments of mostly mild cervical compression [3], and mild compression is age-related. We therefore quantified how far the exploratory pathology classification (main text, Section 3.3) is explained by age.

In the ALL complete-case cohort ($N = 246$; 58 label-positive, 188 controls), labelled subjects were older (median 34.0 vs. 28.0 years; Mann–Whitney $p = 2.0 \times 10^{-7}$). Age alone discriminated the labels with AUC = 0.725, and a linear age+sex score gave AUC = 0.725 (sex alone: 0.53)—above every biomarker-based classifier in the main analysis. To test whether the post-harmonisation improvement in biomarker-based classification merely reflects this age confounding, we residualised each of the eight biomarkers on age and sex (OLS) and repeated the identical leakage-free classification protocol (random forest, 5-fold × 20 repeats, harmonisation refit within each fold):

Table S9. Pathology-label classification before and after residualising the biomarkers on age and sex (leakage-free protocol identical to the main analysis; mean ± SD AUC over 20 repeats).

Features	Original	ComBat
All eight biomarkers (main analysis)	0.520 ± 0.024	0.625 ± 0.018
Biomarkers residualised on age and sex	0.565 ± 0.027	0.655 ± 0.021

The harmonisation-related improvement persists, and in fact slightly increases, after removal of age and sex effects, so it is not explained by the age difference between groups. Nonetheless, because the labels are radiological rather than clinical, are age-related, were not assigned at the extracted vertebral level, and come from a single dataset, we report this endpoint as exploratory and draw no disease-related conclusion from it.

S10. Site- and Scanner-Model-Level Harmonisation

Because only three vendor levels are available, per-metric ComBat was refitted on the HC complete-case sample ($N = 188$) with site (40 levels) or scanner model (12 levels) as the batch variable, using implementations identical to the main analysis. Empirical Bayes shrinkage across batches was enabled for the site- and model-level fits because per-batch samples are small (2–8 subjects per site); for reference, the vendor-batch fit is shown with empirical Bayes both disabled (the canonical per-metric ComBat of the main analysis) and enabled (equivalent to the main-analysis ComBat-joint up to per-feature affine maps). Results:

Table S10. ComBat with alternative batch variables (HC complete-case, $N = 188$; vendor η^2 and site η^2 are means over the eight biomarkers; $r_{\text{pre,post}}$ is the main-text Table 3 metric).

Batch variable	**EB**	**Vendor η^2**	**Site η^2**	**Age R^2**	**Sex R^2**	$r_{\textbf{pre,post}}$
None (Original)	—	0.3147	0.5405	0.0015	0.0298	1.000
Vendor (3 levels; main-analysis ComBat)	off	0.0004	0.2965	0.0089	0.0333	0.789
Vendor (3 levels; ≡ main-analysis ComBat-joint)	on	0.0005	0.2975	0.0087	0.0334	0.806
Scanner model (12 levels)	on	0.0022	0.1972	0.0078	0.0325	0.763
Site (40 levels)	on	0.0120	0.0446	0.0109	0.0490	0.703

Because sites are perfectly nested within vendors, site-level harmonisation also removes vendor effects, down to the chance level ($\eta^2 = 0.012$ vs. the null expectation $(k-1)/(N-1) \approx 0.011$), and it removes most of the site-level variance ($0.54 \rightarrow 0.045$). Scanner-model harmonisation is intermediate. Finer-grained batches recover marginally more age- and sex-related variance but distort subject rankings more (lower $r_{\text{pre,post}}$), and—unlike the vendor-batch fit evaluated in the LOVO framework—a site- or model-level fit offers no route to unseen vendors or unseen sites.